

COMICCRAFT

AI COMIC STORY CREATOR

PROJECT PLANNING & DEVELOPMENT DOCUMENT

PROJECT NAME	ComicCraft - AI Comic Story Creator
TEAM MEMBERS	Dharshini R Indumathi B Nithisha S Swathy D Yogalakshmi J

PROJECT OVERVIEW

ComicCraft is an AI-powered comic story creation platform designed to help creative writers and students transform written story concepts into visual comic strips. The system combines AI-based story generation, image generation, a web interface, and PDF export into a single workflow.

CORE WORKFLOW

Story Prompt → Gemini Story & Dialogue → Panel Scene Prompts → Stable Diffusion Artwork → Web Preview → PDF Export

DOCUMENT CONTENT

Phase	Focus
1	Brainstorming & Idea Prioritization
2	Customer Problem Statement
3	Empathy Map
4	System Architecture & Requirements
5	Development & Testing Plan

PHASE 1: BRAINSTORMING & IDEA PRIORITIZATION

Step 1: Brainstorm and Idea Listing

Each team member listed multiple ideas and suggestions without judgment to address challenges in automated story generation, visual panel creation, and seamless AI comic production.

S.No	Team Member	Idea / Suggestion	Category
1	Dharshini R	Generate comic story outline and dialogue using Gemini AI	Core Feature
2	Indumathi B	Generate comic illustrations using Stable Diffusion API	AI/ML
3	Nithisha S	Develop the backend system using FastAPI	Backend
4	Swathy D	Create a web interface (HTML/CSS) for user inputs	UI/UX
5	Yogalakshmi J	Combine images and text to export as a downloadable PDF	Output

Step 2: Idea Prioritization Matrix

Evaluating candidate features and technical paths based on feasibility and strategic impact for the ComicCraft AI platform.

Final Idea	Feasibility	Importance	Priority	Selected
AI Script & Storyboard Generator (Gemini + Stable Diffusion)	High	High	P1	Yes
FastAPI Web Interface & Panel Previewer	High	High	P1	Yes
Automated PDF Exporter using FPDF	High	Medium	P2	Yes

PHASE 2: CUSTOMER PROBLEM STATEMENT

Understanding user requirements to build an intuitive, end-to-end comic generation experience.

Problem Statement (PS-1)

I am (Customer)	A creative writer or student
I'm trying to	Transform written story concepts into visual comic strips
But	Lacking professional illustration skills and manual design software expertise
Because	Hiring artists is expensive and manual drawing takes significant time
Which makes me feel	Frustrated by inability to visualize and share creative stories effectively

DESIGN GOAL: Provide a simple end-to-end workflow that converts a written idea into structured comic panels and a shareable PDF.

PHASE 3: EMPATHY MAP

Analyzing target user behaviors, motivations, and expectations across four key perspectives to drive customer-centric feature design.

PERSONA: Creative Writer / Student

SAYS

- "I have great story ideas, but I can't draw comic panels."
- "Hiring illustrators is too expensive for small indie stories."

THINKS

- "AI should be able to generate consistent visual scenes from my text."
- "I want to publish my comics online in standard PDF format."

DOES

- Writes scripts and narrative prompts in text editors.
- Searches for affordable graphic generation tools and assets.

FEELS

- Overwhelmed by the complexity of traditional drawing tools.
- Excited by the possibilities of generative AI art.

PHASE 4: SYSTEM ARCHITECTURE & REQUIREMENTS

Core Technology Stack

Component	Technology	Purpose & Description
Backend Framework	FastAPI (Python)	Asynchronous REST API routing, task management, and request processing.
Story Generation AI	Google Gemini	Parsing prompt input into structured multi-panel script & scene prompts.
Image Generation Engine	Stable Diffusion API	Transforming scene text descriptions into comic panel artwork.
Frontend Interface	HTML5, CSS3, Jinja2	Interactive web interface for user prompts, panel views, and edits.
Export Engine	FPDF	Compiling image panels and script text into formatted PDF books.

Directory Architecture Plan

Directory / File	Purpose
app/	Core application backend logic (main.py, routes.py).
templates/	Web UI layout components (index.html).
static/	Generated panel output (panels/), exports (exports/), and styles.
requirements.txt	Python dependencies for local execution.

System Flow

User Prompt → FastAPI → Gemini → Structured Story & Scene Prompts → Stable Diffusion → Panel Preview → FPDF → Downloadable Comic PDF
--

PHASE 5: DEVELOPMENT & TESTING PLAN

Validation & Quality Assurance Strategy

Test Area	Test Objective	Expected Outcome
Script Parsing	Validate Gemini response parsing under varied prompt lengths	Correctly parse prompts containing panel dialogues and visual prompts
Image Generation	Verify Stable Diffusion image output speed and resolution	Consistent visual panel generation
Pipeline Integration	End-to-end user flow test from prompt to panel render	Seamless execution within local Uvicorn environment
PDF Compilation	Export test with multi-page comic layouts	Cleanly formatted PDF ready for sharing

Project Summary & Delivery

ComicCraft AI streamlines digital comic creation by pairing generative text models with diffusion-based rendering. The complete repository source code, environment templates, and documentation are hosted on GitHub for version control and community deployment.

Reference Note

The phase-based organization and project-planning structure were prepared with reference to the uploaded Brainstorming & Ideation document. The ComicCraft-specific content in this PDF follows the material supplied for this project.